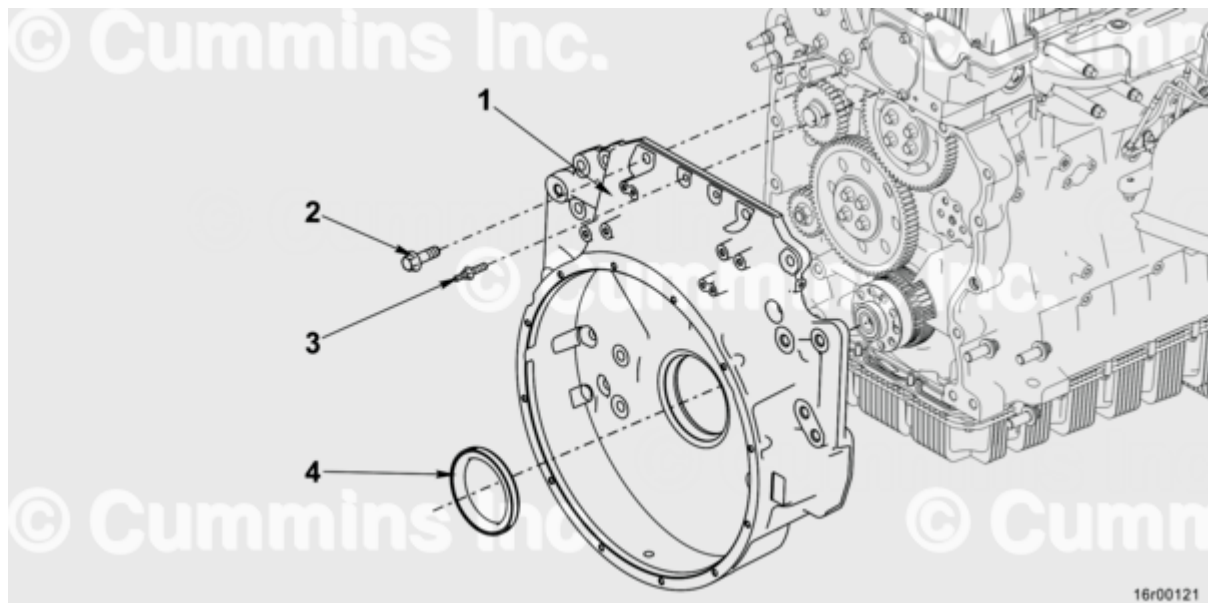


Trainings ▾

Exploded View



Flywheel Housing Exploded View

1. Flywheel housing - quantity 1
2. Capscrew M16 - quantity 16
3. Capscrew M10 - quantity 2
4. Oil seal - quantity 1

Select Service Tools

Recommended Cummins® Service Tools

- Guide pin, Part Number 3376488
- Guide pin, Part Number 3164765
- Abrasive pad, Part Number 3823258
- Loctite® 5188, Part Number 5394294, or equivalent
- Barring tool, Part Number 4919092
- Dial indicator gauge, Part Number 3376050
- Magnetic base, Part Number 3377399

Additional Service Items

- Hoist or lifting fixture
- Rubber hammer
- Pry bar.

Preparatory Steps

WARNING

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

WARNING

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

WARNING

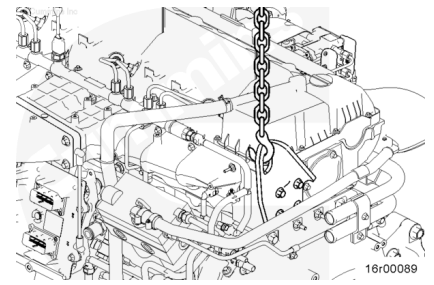
Support the rear of the engine using the rear support attached to the rear of the cylinder block. Failure to support the engine can cause personal injury.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/1016/1016-013-020-tr.html>)
- Remove the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Remove the transmission, clutch, and all related components, if equipped. See equipment manufacturer service information.
- Remove the retarder coolant pipe bracket, if required. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Remove the coolant return pipe for air compressor, if required. Refer to Procedure 012-004 in Section 12. (</qs3/pubsys2/xml/en/procedures/1016/1016-012-004.html>)
- Remove the flywheel assembly. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/1016/1016-016-005.html>)
- Remove any Original Equipment Manufacturer (OEM) attached components (mufflers, shift mechanisms, air filters) from the flywheel housing. See equipment manufacturer service information.

Remove

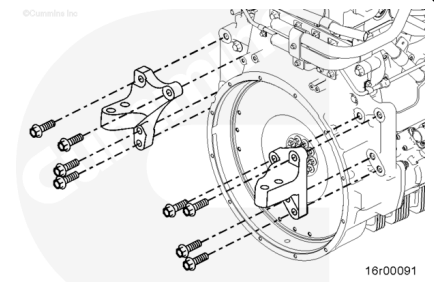
⚠ WARNING ⚠

The engine lifting equipment must be designed to lift the engine and transmission as an assembly without causing personal injury.



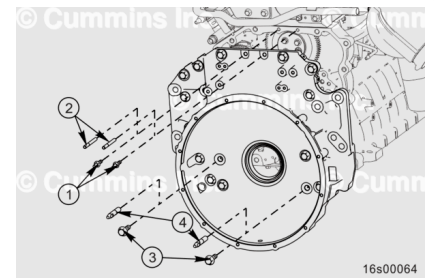
Use a hoist or lifting fixture to support the rear of the engine.

Remove the rear engine supports. Refer to Procedure 016-003 in Section 16. (</qs3/pubsys2/xml/en/procedures/1016/1016-016-003.html>)



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.



Remove the M10 capscrews (1) and two M16 capscrews (3).

Install the M10 guide pins (2), Part Number 3376488, and M16 guide pins (4), Part Number 3164765, into the flywheel housing.

Loosen the flywheel housing capscrews, but do **not** remove.

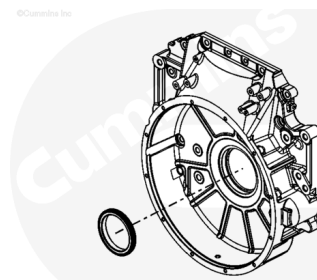
Use a rubber hammer to loosen the flywheel housing so that the seal is broken between the flywheel housing and rear gear housing.

While supporting the flywheel housing, remove the mounting capscrews and the flywheel housing.

Note : Make sure to note the position of any locating dowel rings.

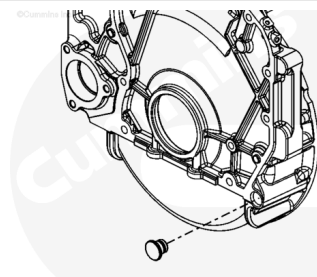
Disassemble

Remove the rear crankshaft seal.



16r00097

Remove the plug from the barring gear hole. Discard the o-ring.



16r00098

Clean and Inspect for Reuse

⚠ WARNING ⚠

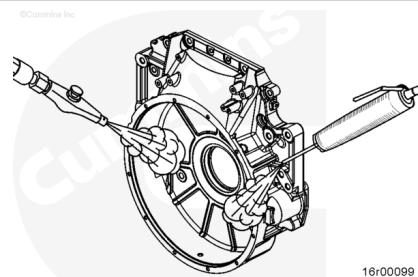
When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

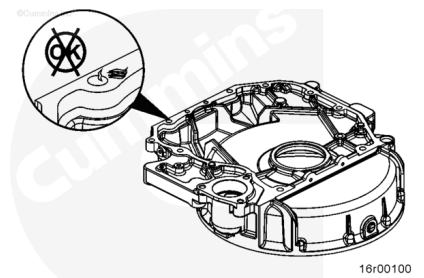


16r00099

Use steam or solvent to clean the flywheel housing.

Dry with compressed air.

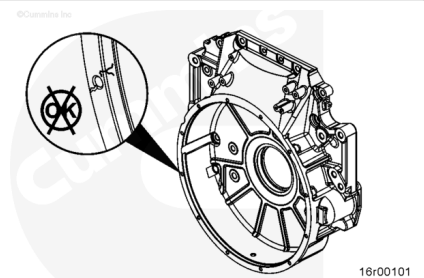
Inspect the flywheel housing for cracks, especially in the area of the flywheel housing that mounts to the cylinder block or rear gear housing.



Inspect the flywheel housing transmission/drive unit mounting surface for cracks.

Also inspect for damaged threads commonly caused by cross-threaded capscrews or installing an incorrect capscrew.

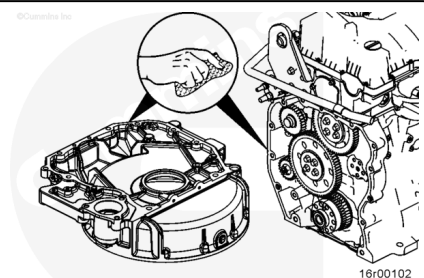
Note : Helicoils are available to repair damaged threads.



Inspect the rear face of the gear housing and flywheel housing mounting surface for cleanliness and raised nicks or burrs.

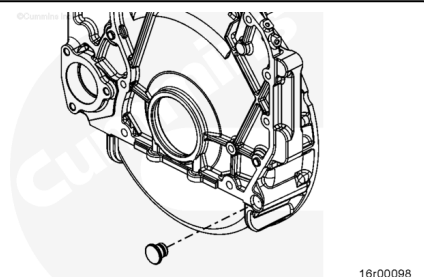
Use an abrasive pad, Part Number 3823258, to remove small nicks and burrs.

Thoroughly clean the flywheel housing and gear housing mating surfaces. These surfaces **must** be clean of oil and debris.



Assemble

Install the barring gear hole plug. Use a new o-ring.



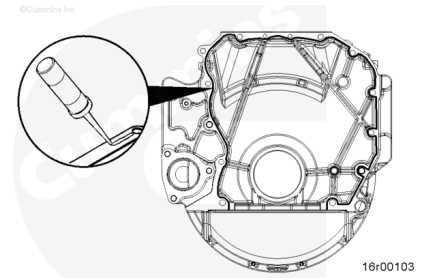
Install

Note : If the flywheel housing (or rear gear housing for Rear Gear Train (RGT) engines) is being replaced or if troubleshooting a vibration/alignment issue, see the Measure section of this procedure below. Alternative steps must be performed before installing a new flywheel housing or if the dowel rings are removed during a previous repair.

Ensure all mating surfaces are cleaned and free from debris, lubrication, and preservative.

Apply a 2 to 3 mm silicon bead of Loctite® 5188, Part Number 5394294, or equivalent, to the rear flywheel mounting surface, as shown.

Note : Make sure silcon bead path is close to the lower edge as indicated in image.



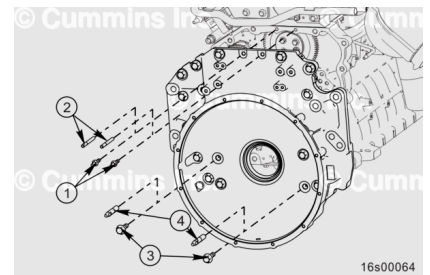
Note : Before installing the flywheel housing, make sure any locating dowel rings are in the same position as when the flywheel housing is removed. If the locating dowel rings are found to be damaged after a visual inspection, replace the dowel rings.

Install the flywheel housing and M16 capscrews.

Tighten the flywheel housing capscrews finger-tight.

Remove the guide pins (2 and 4).

Install M10 capscrews (1) and remaining M16 capscrews (3) finger-tight.



See the illustration for the flywheel housing capscrew torque sequence.

Note : Use clean lubricating engine oil to lightly coat the capscrew threads before installation.

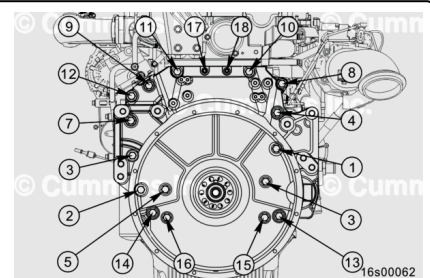
Torque Value:

M16 (1-16)

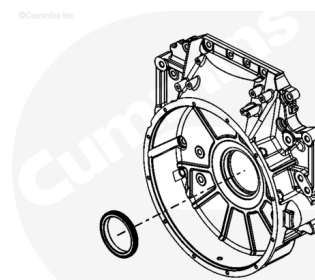
1. 100 n•m [74 ft-lb]
2. 217 n•m [160 ft-lb]

Torque Value:

M10 (17, 18) 65 n•m [48 ft-lb]



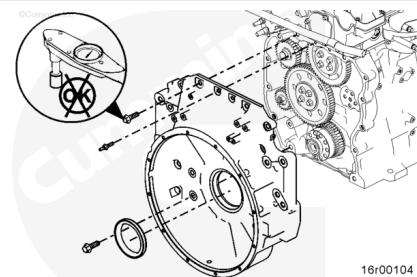
Install a new rear crankshaft seal. Refer to Procedure 001-024 in Section 1. (/qs3/pubsys2/xml/en/procedures/1016/1016-001-024.html)



Measure

⚠ CAUTION ⚠

When barring the engine using barring tool, Part Number 4919092, be careful not to apply excessive side loading to the flywheel housing. This may cause the flywheel housing to move and cause inaccurate measurement readings.

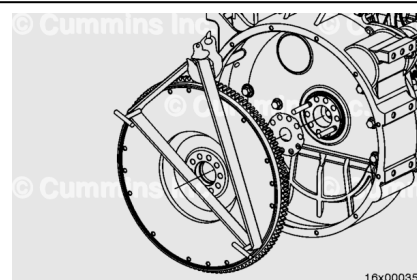


Note : Follow this step **only** if troubleshooting a vibration/alignment issue.

Install the flywheel housing following the Install section of this procedure, but do **not** tighten the capscrews. **Only** tighten the capscrews enough to hold the flywheel wheel in place.

Note : For RGT engines, when installing a new flywheel housing to check flywheel housing bore alignment and face alignment, do **not** apply sealant to the flywheel housing prior to installing for measurement.

- Install the flywheel and inspect the flywheel face and bore runout. Refer to Procedure 016-005 in Section 16. (/qs3/pubsys2/xml/en/procedures/1016/1016-016-005.html)
- If the flywheel is within specifications, proceed with measuring the flywheel housing alignment.



Face Alignment

⚠ CAUTION ⚠

The dial indicator tip must not enter the capscrew holes, or the gauge will be damaged.

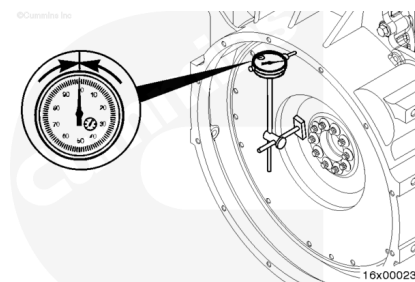
Face alignment is determined by calculating the total indicator reading.

Attach the dial indicator gauge, Part Number 3376050, mounted by a magnetic base, Part Number 3377399, to the flywheel as illustrated.

Note : The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid, so it does not sag. If the bar sags, the magnetic base moves, or the indicator slips, the readings obtained will not be accurate.

Position the indicator at the 12 o'clock position, and zero the gauge.

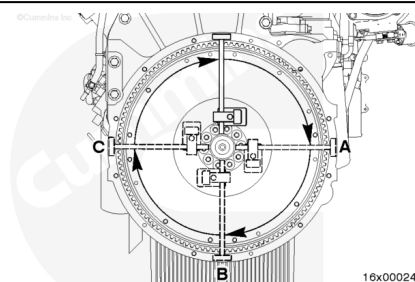
Note : The crankshaft must be pushed toward the front of the engine to remove the crankshaft end clearance each time a position is measured.



Use the barring tool, Part Number 4919092, to slowly rotate the crankshaft.

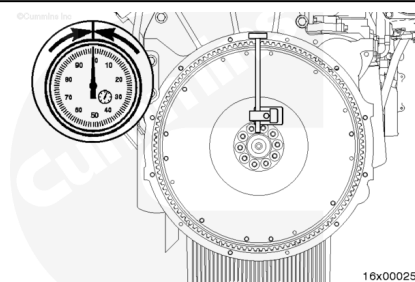
Record the readings at the 3 o'clock (A), 6 o'clock (B), and 9 o'clock (C) positions.

The values for A, B, and C can be positive or negative.



Continue to rotate the crankshaft until the indicator is at the 12 o'clock position.

Check the indicator to make sure the needle points to zero. If it does **not**, the readings will be incorrect and the procedure will have to be completed again.



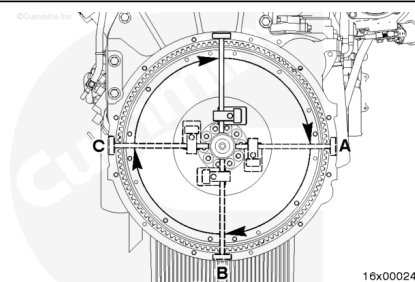
Determine the total indicator reading.

Total indicator reading is determined by calculating the difference between the highest and lowest measurement from the four locations measured.

As the example below illustrates, the total indicator reading will be:

$$+0.08 \text{ mm} - (-0.05 \text{ mm}) = 0.13 \text{ mm}$$

$$[+0.003 \text{ in} - (-0.002 \text{ in}) = 0.005 \text{ in}]$$

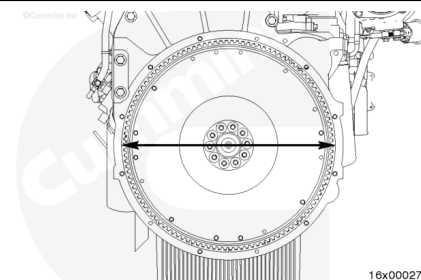


Example:

12 o'clock	0.00 mm	[0.000 in]
3 o'clock	+0.08 mm	[+0.003 in]
6 o'clock	-0.05 mm	[-0.002 in]
9 o'clock	+0.08 mm	[+0.003 in]
Equals total indicator reading	0.13 mm	[0.005 in]

The maximum allowable total indicator reading is determined by the diameter of the housing bore. If out of specifications, replace the housing.

Note : For RGT engines, the rear gear housing may also be the cause of the total indicator reading being out of specification.

**Flywheel Housing Bore and Face Runout**

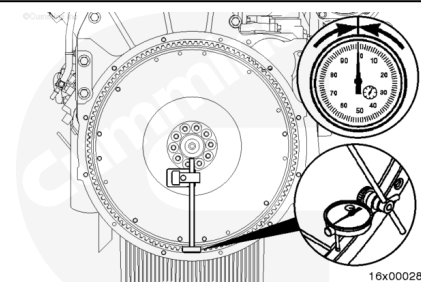
Society of Automotive Engineers (SAE) Number	Bore Diameter		Total Indicator Reading Maximum	
	mm	in	mm	in
1	510.98 to 511.38	20.117 to 20.133	0.36	0.014

Bore Alignment

Attach the dial indicator gauge, Part Number 3376050, mounted by a magnetic base, Part Number 3377399, to the flywheel as illustrated.

Note : The dial indicator can be mounted by any method that holds the extension bar of the indicator rigid, so it does not sag. If the bar sags, the magnetic base moves, or the indicator slips, the readings obtained will not be accurate.

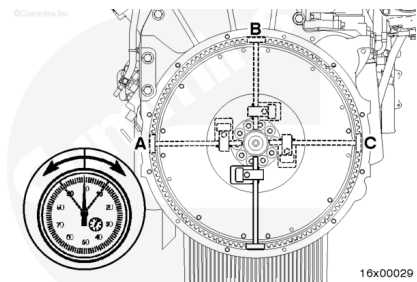
Position the indicator in the 6 o'clock position, and zero the gauge.



Slowly rotate the crankshaft. Record the readings obtained at the 9 o'clock, 12 o'clock, and 3 o'clock positions as A, B, and C in the concentricity worksheet.

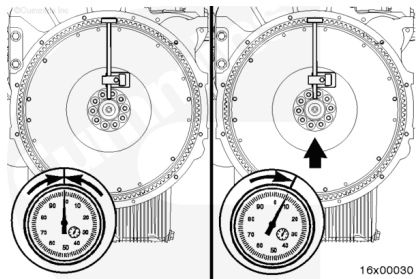
Recheck zero at the 6 o'clock position. If it does **not**, the readings will be incorrect and the procedure will have to be completed again.

The values for A, B, and C can be positive or negative. See the illustration to determine the correct sign when recording these values.



⚠ CAUTION ⚠

Do not force the crankshaft beyond the point where the bearing clearance has been removed. Do not pry against the flywheel housing. These actions can cause false bearing clearance readings and result in engine damage.



Rotate the crankshaft until the dial indicator is at the 12 o'clock position and zero the gauge.

Use a pry bar to raise the rear of the crankshaft to its upper limit. Record the value as D on the concentricity worksheet. This is the vertical bearing clearance adjustment, which will **always** be positive.

Create a concentricity worksheet, as illustrated, to determine the values for the "total vertical" and "total horizontal" values.

Note : The values listed in the concentricity worksheet illustrated are for example only and are listed in inches. The actual numbers measured may differ.

Input the values recorded for A, B, C, and D into the concentricity worksheet.

The total horizontal is the 9 o'clock reading (A) minus the 3 o'clock reading (C).

The total vertical is the 12 o'clock reading (B) plus the bearing clearance (D).

Example:

- 6 o'clock = reference = 0
- 9 o'clock = (A) = 0.004
- 12 o'clock = (B) = 0.003
- 3 o'clock = (C) = (-0.002)

Concentricity Worksheet	
9 o'clock	a = 0.004
3 o'clock	c = -0.002
Total Horizontal	a - c = .006
12 o'clock	b = .003
Bearing Clearance	d = .002
Total Vertical	b + d = .005

Use the worksheet and the numbers from the example, the total horizontal value equals 0.006 and the total vertical value equals 0.005.

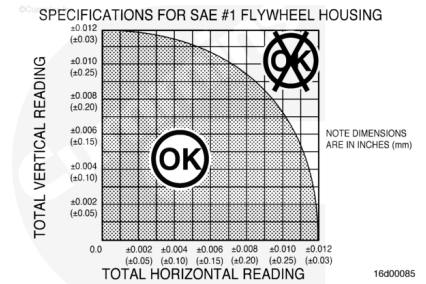
Note : Use the corresponding chart for the SAE 1 flywheel housing being measured.

Use the illustration, mark the total horizontal value on the horizontal side of the chart and the total vertical on the vertical side of the chart.

Use a straightedge to find the intersection point of the total horizontal and total vertical values. The intersection point **must** fall within the shaded area for the flywheel housing concentricity to be within specification.

Use the total horizontal and total vertical values from the previous example, the intersection point falls within the shaded area. Therefore, the flywheel housing concentricity is within specification.

Note : Make sure to use the correct total indicator reading specifications for the flywheel housing being measured when comparing measurements.



Tighten the capscrews in the sequence shown.

Note : Use clean lubricating engine oil to lightly coat the capscrew threads before installation.

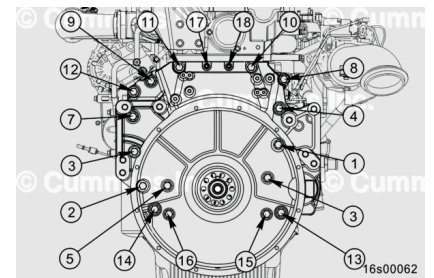
Torque Value:

M16 (1-16)

1. 100 n•m [74 ft-lb]
2. 217 n•m [160 ft-lb]

Torque Value:

M10 (17, 18) 65 n•m [48 ft-lb]



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the rear engine support brackets. Refer to Procedure 016-003 in Section 16. (</qs3/pubsys2/xml/en/procedures/1016/1016-016-003.html>)
- Install the flywheel assembly. Refer to Procedure 016-005 in Section 16. (</qs3/pubsys2/xml/en/procedures/1016/1016-016-005.html>)
- Install the starting motor. Refer to Procedure 013-020 in Section 13. (</qs3/pubsys2/xml/en/procedures/1016/1016-013-020-tr.html>)
- Install the oil pan. Refer to Procedure 007-025 in Section 7. (</qs3/pubsys2/xml/en/procedures/1016/1016-007-025.html>)
- Install the coolant return pipe for air compressor, if removed. Refer to Procedure 012-004 in Section 12. (</qs3/pubsys2/xml/en/procedures/1016/1016-012-004.html>)
- Install the retarder coolant pipe bracket, if removed. Refer to Procedure 008-152 in Section 8. (</qs3/pubsys2/xml/en/procedures/1016/1016-008-152.html>)
- Install the transmission and related components, if equipped. See equipment manufacturer service information.
- If previously removed, attached any OEM components (mufflers, shift mechanisms, air filters) to the flywheel housing. See equipment manufacturer service information.
- Connect the batteries. See equipment manufacturer service information.
- Operate the engine and check for leaks.